

SRPT-LAS Proof Sketch

Bryan Yuen and Izzy Grosz

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1 Introduction

A key component of scheduling models in queueing is the job size information available to the scheduler. The vast majority of models assume either known-size jobs, unknown-size jobs, or known job size distributions, giving rise to policies such as First-Come First-Served (FCFS), Shortest Remaining Processing Time (SRPT), and the Gittins index policy. However, practical systems often contain a mixture of accurate and inaccurate predictions.

In this work, we focus on the mixed-knowledge model in an $M/G/1$ queue, an under-studied setting where job sizes are either exactly known or completely unknown. To minimize the mean response time in this environment, we propose the SRPT-LAS policy.

SRPT-LAS is a SOAP scheduling policy that ranks known-size jobs by their remaining processing time and unknown-size jobs by their age.

This document provides a formal proof sketch of our main optimality result: as the fraction of known-size jobs, denoted as p_k , tends to 1, the mean response time of SRPT-LAS converges to that of SRPT.

To prove this result, we utilize a tagged job approach to characterize the difference in the mean response time between our SRPT-LAS system and the standard SRPT system. As detailed in the subsequent lemmas, we decompose this delay difference for a job of size x into three specific components.

By evaluating the $p_k \rightarrow 1$ limit for each component independently, we demonstrate that all three differences vanish, thereby completing the proof.

2 Defining SRPT-LAS

To manage a mixed-knowledge $M/G/1$ system, we propose a hybrid scheduling policy named SRPT-LAS. The core mechanism of this policy is that it always serves the job in the system with the lowest rank.

We can formally define this ranking mechanism. Let a job in the system have an original size x (if known) and an attained service (or age) a .

Definition 1. Under the SRPT-LAS policy, the rank of a job is defined according to its information type:

- **Known-Size Jobs (Type k):** The rank is the job's remaining processing time, calculated as $x - a$, which behaves exactly as it would in the SRPT policy.
- **Unknown-Size Jobs (Type u):** The rank is the job's age, a , which behaves exactly as it would in the LAS policy.

At any given moment, SRPT-LAS preemptively serves the available job that has the minimum rank.

3 Defining Mixed-Knowledge M/G/1

In our M/G/1 queueing system: jobs arrive according to a Poisson process with rate λ . Each job is one of two types, with different availability of size information:

Known-Size Jobs (Type k): With probability p_k i.i.d., the job's exact size is known to the scheduler on arrival. The size of these jobs is sampled i.i.d. from a distribution with random variable S_k .

Unknown-Size Jobs (Type u): Otherwise, a job's size is entirely unknown upon arrival. The size of these jobs is sampled i.i.d. from a distribution with random variable S_u .

The size of a general arrival to the system is denoted by the random variable S . We allow the scheduling policy to serve any job at any time, with preempt-resume service.

4 Defining limit p_k

Definition 2. In a mixed-knowledge system, the overall job size S is a mixture distribution dependent on the probability of a job being known (p_k), the distribution of known sizes (S_k), and the distribution of unknown sizes (S_u). We require the overall arrival rate λ and the overall size distribution S to remain fixed. We allow any combination of (p_k, S_k, S_u) triples that preserve a constant overall distribution S in the $p_k \rightarrow 1$ limit.

To formalize this, we define the limit through a sequence of systems.

A *Distributional Sequence* (DS) is a sequence of system parameters indexed by i , denoted as $\{p_k^{(i)}, S_k^{(i)}, S_u^{(i)}\}_{i=1}^{\infty}$, that satisfies the two properties. First, we define the overall job size distribution $S^{(i)}$ to be drawn from $S_k^{(i)}$ with probability $p_k^{(i)}$ and from $S_u^{(i)}$ with probability $1 - p_k^{(i)}$, i.i.d.

Now, we can define the two key properties of the sequence:

1. For all i , the overall job size distribution $S^{(i)}$ remains constant. As a result, we omit the superscript, calling the overall distribution S .
2. As $i \rightarrow \infty$, the probability of a job arriving with a known size approaches 1: $\lim_{i \rightarrow \infty} p_k^{(i)} = 1$.

We assume standard stability conditions ($\rho < 1$) hold for all systems in the sequence.

Using this distributional sequence, proving that the mean response time of SRPT-LAS converges to SRPT as $p_k \rightarrow 1$ is equivalent to proving that for any valid DS, the expected delay difference between the two policies approaches 0 as $i \rightarrow \infty$.

5 Main Convergence Result

5.1 Decomposition Lemmas

Lemma 1. *The overall expected response time under SRPT-LAS is given by:*

$$E[T^{SL}] = \int_{x=0}^{\infty} E[T_x^{SL}] f_S(x) dx \quad (1)$$

where the expected response time for a job of size x under SRPT-LAS is:

$$E[T_x^{SL}] = p_k(x) E[T_{k,x}^{SL}] + (1 - p_k(x)) E[T_{u,x}^{SL}] \quad (2)$$

The probability $p_k(x)$ that a job of size x is of type k is:

$$p_k(x) = p(\text{type} = k \mid S = x) = \frac{p_k f_{S_k}(x)}{p_k f_{S_k}(x) + p_u f_{S_u}(x)} \quad (3)$$

where $p_u = 1 - p_k$. The expected response times for known and unknown tagged jobs of size x are defined as:

$$E[T_{k,x}^{SL}] = \frac{\lambda E[S_{\bar{x}}^2]}{2(1 - \rho_{\bar{x}})^2} + \int_{t=0}^x \frac{dt}{1 - \rho_t} \quad (4)$$

$$E[T_{u,x}^{SL}] = \frac{\lambda E[S_{\bar{x}}^2]}{2(1 - \rho_{\bar{x}})^2} + \frac{x}{1 - \rho_{\bar{x}}} \quad (5)$$

where:

$$\rho_{\bar{x}} = \lambda E[S_{\bar{x}}] = \lambda (p_k E[S_{k(\leq x)}] + p_u E[S_{u(\bar{x})}]) \quad (6)$$

with $S_{k(\leq x)} = S_k \mathbb{1}(S_k \leq x)$ and $S_{u(\bar{x})} = \min(S_u, x)$.

Proof idea. Using a tagged job analysis with two different types of jobs. $\square$

Lemma 2. In any mixed-knowledge $M/G/1$, for a tagged job with size x ,

$$E[T_x^{SL}] - E[T_x^{SRPT}] = \Delta \text{Wait}(x) + \Delta \text{Res}_1(x) + \Delta \text{Res}_2(x),$$

where

$$\Delta \text{Wait}(x) = \frac{\lambda E[S_{\bar{x}}^2]}{2} \left(\frac{1}{(1 - \rho_{\bar{x}})^2} - \frac{1}{(1 - \rho_x)^2} \right) \quad (7)$$

$$\Delta \text{Res}_1(x) = p_k(x) \left(\int_{t=0}^x \frac{dt}{1 - \rho_t} - \int_{t=0}^x \frac{dt}{1 - \rho_t} \right) \quad (8)$$

$$\Delta \text{Res}_2(x) = (1 - p_k(x)) \left(\frac{x}{1 - \rho_{\bar{x}}} - \int_{t=0}^x \frac{dt}{1 - \rho_t} \right) \quad (9)$$

Proof.

$$\begin{aligned} E[T_x^{SL}] - E[T_x^{SRPT}] &= \left(\frac{\lambda E[S_{\bar{x}}^2]}{2(1 - \rho_{\bar{x}})^2} + (1 - p_k(x)) \frac{x}{1 - \rho_{\bar{x}}} + p_k(x) \int_{t=0}^x \frac{dt}{1 - \rho_t} \right) - \left(\frac{\lambda E[S_{\bar{x}}^2]}{2(1 - \rho_x)^2} + \int_{t=0}^x \frac{dt}{1 - \rho_t} \right) \\ &= \left(\frac{\lambda E[S_{\bar{x}}^2]}{2(1 - \rho_{\bar{x}})^2} - \frac{\lambda E[S_{\bar{x}}^2]}{2(1 - \rho_x)^2} \right) + \left(p_k(x) \int_{t=0}^x \frac{dt}{1 - \rho_t} - p_k(x) \int_{t=0}^x \frac{dt}{1 - \rho_t} \right) \\ &\quad + \left((1 - p_k(x)) \frac{x}{1 - \rho_{\bar{x}}} - (1 - p_k(x)) \int_{t=0}^x \frac{dt}{1 - \rho_t} \right) \\ &= \frac{\lambda E[S_{\bar{x}}^2]}{2} \left(\frac{1}{(1 - \rho_{\bar{x}})^2} - \frac{1}{(1 - \rho_x)^2} \right) + p_k(x) \left(\int_{t=0}^x \frac{dt}{1 - \rho_t} - \int_{t=0}^x \frac{dt}{1 - \rho_t} \right) \\ &\quad + (1 - p_k(x)) \left(\frac{x}{1 - \rho_{\bar{x}}} - \int_{t=0}^x \frac{dt}{1 - \rho_t} \right) \\ &= \Delta \text{Wait}(x) + \Delta \text{Res}_1(x) + \Delta \text{Res}_2(x) \quad \square \end{aligned}$$

Preliminary identity: partial load difference

$$\begin{aligned}
\rho_{\dot{x}} &= \lambda E[S_{\dot{x}}] = \lambda (p_k E[S_{k(x)}] + p_u E[S_{u(\bar{x})}]) \\
\rho_x &= \lambda E[S_x] \\
\rho_{\dot{x}} &= \lambda p_k E[S_{k(x)}] + \lambda p_u E[S_{u(\bar{x})}] \\
\rho_{\dot{x}} &= \lambda p_k E[S_{k(x)}] + \lambda p_u E[S_{u(x)}] + \lambda p_u x P(S_u > x) \\
\rho_x &= \lambda E[S_x] = \lambda p_k E[S_{k(x)}] + \lambda p_u E[S_{u(x)}] \\
\rho_{\dot{x}} &= \rho_x + \lambda p_u x P(S_u > x) \\
\rho_{\dot{x}} - \rho_x &= \lambda p_u x P(S_u > x)
\end{aligned}$$

Preliminary identity: derivatives of ρ_x and $\rho_{\bar{x}}$

$$\begin{aligned}
\rho_x &= \lambda E[S \mathbb{I}(S \leq x)] \\
\frac{d}{dx} \rho_x &= \lambda x f_S(x) \\
\rho_{\bar{x}} &= \lambda E[\min(S, x)] \\
\rho_{\bar{x}} &= \rho_x + \lambda x P(S \geq x) \\
\frac{d}{dx} \rho_{\bar{x}} &= \lambda x f_S(x) - \lambda x f_S(x) + \lambda P(S \geq x) \\
&= \lambda P(S \geq x)
\end{aligned}$$

5.2 Waiting time difference $\Delta \text{Wait}(x)$

Lemma 3. *In any mixed-knowledge $M/G/1$, for any distributional sequence, and for any tagged job size x ,*

$$\lim_{p_k \rightarrow 1} \Delta \text{Wait}(x) = 0$$

Proof.

$$\begin{aligned}
\frac{1}{(1 - \rho_{\dot{x}})^2} - \frac{1}{(1 - \rho_x)^2} &= \frac{(1 - \rho_x)^2 - (1 - \rho_{\dot{x}})^2}{(1 - \rho_{\dot{x}})^2 (1 - \rho_x)^2} \\
&= \frac{1 - 2\rho_x + (\rho_x)^2 - 1 + 2\rho_{\dot{x}} - (\rho_{\dot{x}})^2}{(1 - \rho_{\dot{x}})^2 (1 - \rho_x)^2} \\
&= \frac{(\rho_x)^2 - (\rho_{\dot{x}})^2 + 2\rho_{\dot{x}} - 2\rho_x}{(1 - \rho_{\dot{x}})^2 (1 - \rho_x)^2} \\
&= \frac{(\rho_{\dot{x}} - \rho_x)(2 - \rho_{\dot{x}} - \rho_x)}{(1 - \rho_{\dot{x}})^2 (1 - \rho_x)^2} \\
\lim_{p_k \rightarrow 1} \left(\frac{1}{(1 - \rho_{\dot{x}})^2} - \frac{1}{(1 - \rho_x)^2} \right) &= \lim_{p_u \rightarrow 0} \left(\frac{1}{(1 - \rho_{\dot{x}})^2} - \frac{1}{(1 - \rho_x)^2} \right) \\
&= \lim_{p_u \rightarrow 0} \frac{(\rho_{\dot{x}} - \rho_x)(2 - \rho_{\dot{x}} - \rho_x)}{(1 - \rho_{\dot{x}})^2 (1 - \rho_x)^2} \\
&= \lim_{p_u \rightarrow 0} \frac{(\lambda p_u x P(S_u > x))(2 - \rho_{\dot{x}} - \rho_x)}{(1 - \rho_{\dot{x}})^2 (1 - \rho_x)^2} \\
&= 0
\end{aligned}$$

$$\lim_{p_k \rightarrow 1} \Delta \text{Wait}(x) = \frac{\lambda E[S_{\bar{x}}^2]}{2} \left(\frac{1}{(1 - \rho_{\bar{x}})^2} - \frac{1}{(1 - \rho_x)^2} \right) = \frac{\lambda E[S_{\bar{x}}^2]}{2} \lim_{p_k \rightarrow 1} \left(\frac{1}{(1 - \rho_{\bar{x}})^2} - \frac{1}{(1 - \rho_x)^2} \right) = 0$$

□

5.3 First Residence Time Difference ($\Delta \text{Res}_1(x)$)

Lemma 4. *In any mixed-knowledge $M/G/1$, for any distributional sequence, and for any tagged job size x ,*

$$\lim_{p_k \rightarrow 1} \Delta \text{Res}_1(x) = 0$$

Proof.

$$\begin{aligned} \Delta \text{Res}_1(x) &= p_k(x) \left(\int_{t=0}^x \frac{dt}{1 - \rho_i} - \int_{t=0}^x \frac{dt}{1 - \rho_t} \right) \\ &= p_k(x) \int_{t=0}^x \left(\frac{1}{1 - \rho_i} - \frac{1}{1 - \rho_t} \right) dt \\ &= p_k(x) \int_{t=0}^x \frac{(1 - \rho_t) - (1 - \rho_i)}{(1 - \rho_i)(1 - \rho_t)} dt \\ &= p_k(x) \int_{t=0}^x \frac{\rho_i - \rho_t}{(1 - \rho_i)(1 - \rho_t)} dt \end{aligned}$$

From our previous analysis on the waiting time difference, we established that for any size boundary t :

$$\rho_i - \rho_t = \lambda p_u t P(S_u > t)$$

Substituting this relationship into the integral gives:

$$\Delta \text{Res}_1(x) = p_k(x) \int_{t=0}^x \frac{\lambda p_u t P(S_u > t)}{(1 - \rho_i)(1 - \rho_t)} dt$$

$$\lim_{p_u \rightarrow 0} p_k(x) = \lim_{p_u \rightarrow 0} \frac{p_k f_{S_k}(x)}{p_k f_{S_k}(x) + p_u f_{S_u}(x)} = \frac{1 \cdot f_{S_k}(x)}{1 \cdot f_{S_k}(x) + 0} = 1$$

$$\begin{aligned} \lim_{p_k \rightarrow 1} \Delta \text{Res}_1(x) &= \left(\lim_{p_u \rightarrow 0} p_k(x) \right) \cdot \lim_{p_u \rightarrow 0} \int_{t=0}^x \frac{\lambda p_u t P(S_u > t)}{(1 - \rho_i)(1 - \rho_t)} dt \\ &= 1 \cdot \lim_{p_u \rightarrow 0} p_u \left(\int_{t=0}^x \frac{\lambda t P(S_u > t)}{(1 - \rho_i)(1 - \rho_t)} dt \right) \\ &= 1 \cdot 0 \cdot \left(\lim_{p_u \rightarrow 0} \int_{t=0}^x \frac{\lambda t P(S_u > t)}{(1 - \rho_t)^2} dt \right) \end{aligned}$$

We know that

$$\lim_{p_u \rightarrow 0} \int_{t=0}^x \frac{\lambda t P(S_u > t)}{(1 - \rho_t)^2} dt \leq \int_{t=0}^x \frac{\lambda t}{(1 - \rho_t)^2} dt \leq \infty$$

as

$$\int_{t=0}^x \frac{\lambda t}{(1 - \rho_t)^2} dt$$

doesn't depend on p_u .

Therefore,

$$\lim_{p_k \rightarrow 1} \Delta \text{Res}_1(x) = 0$$

□

5.4 Second Residence Time Difference ($\Delta \text{Res}_2(x)$)

[FAILED]

Lemma 5. *In any mixed-knowledge $M/G/1$, for any distributional sequence, and for any tagged job size x ,*

$$\lim_{P_k \rightarrow 1} \Delta \text{Res}_2(x) = 0$$

Proof.

$$\Delta \text{Res}_2(x) = (1 - p_k(x)) \left(\frac{x}{1 - \rho_{\dot{x}}} - \int_{t=0}^x \frac{dt}{1 - \rho_t} \right)$$

We evaluate the limit of this term as the fraction of unknown jobs approaches zero ($p_u \rightarrow 0$, which implies $p_k \rightarrow 1$). From our earlier evaluation of the known job probability density, we established:

$$\lim_{p_u \rightarrow 0} p_k(x) = \lim_{p_u \rightarrow 0} \frac{p_k f_{S_k}(x)}{p_k f_{S_k}(x) + p_u f_{S_u}(x)} = 1 \quad (10)$$

Therefore, the limit of the first factor is $\lim_{p_u \rightarrow 0} (1 - p_k(x)) = 0$. (FALSE!)

Next, we examine the delay factor inside the parentheses. Under our assumed stability condition for the system ($\rho < 1$), the partial loads $\rho_{\dot{x}}$ and ρ_t are strictly bounded below 1 for all valid job sizes. The terms evaluate to a finite value:

$$\frac{x}{1 - \rho_{\dot{x}}} - \int_{t=0}^x \frac{dt}{1 - \rho_t} < \infty \quad (11)$$

Since the limit of a product is the product of the limits, and the second factor is finitely bounded, the entire term vanishes:

$$\lim_{p_k \rightarrow 1} \Delta \text{Res}_2(x) = 0 \cdot (\text{finite value}) = 0 \quad (12)$$

□

Remark: a failed first attempt

This statement is false! Counterexample:

$$S_u = \text{Uniform}\left(\left[\frac{1 - p_u}{2}, \frac{1 + p_u}{2}\right]\right), S_k = \text{Uniform}\left(\left[0, \frac{1 - p_2}{2}\right] \cup \left[\frac{1 + p_u}{2}, 1\right]\right)$$

Note: $1/2$ can be replaced by any x .

6 Convergence rate results

6.1 Moment Conditions and Integrability

Expected Response Time Differences

The fundamental quantities we are comparing:

$$\begin{aligned} \mathbb{E}[T^{SL}] - \mathbb{E}[T^{SRPT}] \\ \mathbb{E}[T_{(x)}^{SL}] - \mathbb{E}[T_{(x)}^{SRPT}] \end{aligned}$$

Evaluating Limits Inside Integrals

To prove the expected differences converge to 0, we must justify passing the limit inside the integral (likely via the Dominated Convergence Theorem):

$$\begin{aligned} & \lim_{p_u \rightarrow 0} \int g(x, p_u) \\ & \text{If } \forall x, \lim_{p_u \rightarrow 0} g(x, p_u) = 0 \\ \implies & \lim_{p_u \rightarrow 0} \int g(x, p_u) = 0 ? \end{aligned}$$

6.2 Bounding Expected Wait Time Difference

$$\begin{aligned} \Delta Wait(x) &= \frac{\lambda E[S_{\bar{x}}^2]}{2} \left(\frac{1}{(1 - \rho_{\dot{x}})^2} - \frac{1}{(1 - \rho_x)^2} \right) \\ \frac{1}{(1 - \rho_{\dot{x}})^2} - \frac{1}{(1 - \rho_x)^2} &= \frac{(\rho_{\dot{x}} - \rho_x)(2 - \rho_{\dot{x}} - \rho_x)}{(1 - \rho_{\dot{x}})^2(1 - \rho_x)^2} \end{aligned}$$

Using the bounds $\rho_x \leq \rho$ and $\rho_{\dot{x}} \leq \rho$:

$$\begin{aligned} \frac{1}{(1 - \rho_{\dot{x}})^2(1 - \rho_x)^2} &\leq \frac{1}{(1 - \rho)^4} \\ \rho_{\dot{x}} - \rho_x &= \lambda p_u x P(S_u \geq x) \leq \lambda p_u x \end{aligned}$$

Substituting these bounds back into the factored fraction:

$$\leq \frac{\lambda p_u x \cdot 2}{(1 - \rho)^4}$$

$$\begin{aligned} \Delta Wait(x) &\leq \frac{\lambda E[S_{\bar{x}}^2]}{2} \left(\frac{\lambda p_u x \cdot 2}{(1 - \rho)^4} \right) = p_u \left(\frac{\lambda^2 E[S_{\bar{x}}^2] x}{(1 - \rho)^4} \right) \\ \mathbb{E}[\Delta Wait(S)] &= \int_0^\infty \Delta Wait(x) f_S(x) dx \leq \frac{p_u \lambda^2}{(1 - \rho)^4} \int_0^\infty x E[\min(S, x)^2] f_S(x) dx \end{aligned}$$

$$\begin{aligned}
\int x E[S_x^2] f_S(x) dx &= \int_{x=0}^{\infty} x \int_{y=0}^{\infty} \min(x, y)^2 f_S(y) f_S(x) dy dx \\
&\leq \int_{x=0}^{\infty} \int_{y=0}^{\infty} x^{3/2} y^{3/2} f_S(y) f_S(x) dy dx \\
&= E[S^{3/2}]^2
\end{aligned}$$

$$\int_0^{\infty} x E[\min(S, x)^2] f_S(x) dx \leq (E[S^{3/2}])^2,$$

Therefore

$$\mathbb{E}[\Delta Wait(S)] \leq p_u \frac{\lambda^2 (E[S^{3/2}])^2}{(1-\rho)^4}.$$

Theorem: $\forall \lambda, \forall S$ with $E[S^{3/2}] < \infty$, $\exists C_{\lambda, S}$ such that $\forall (P_k, S_k, S_u)$ sequence, $\mathbb{E}[\Delta Wait(S)] \leq p_u C_{\lambda, S}$.

Proof: $C_{\lambda, S} = \frac{\lambda^2 (E[S^{3/2}])^2}{(1-\rho)^4}.$

6.3 Bounding Expected First Residence Time Difference

$$\begin{aligned}
\int_{t=0}^x \frac{\lambda t P(S_u > t)}{(1-\rho_t)^2} dt &\leq \int_{t=0}^x \frac{\lambda t}{(1-\rho_t)^2} dt \\
&\leq \frac{\lambda x}{(1-\rho)^2} \\
\Delta Res_1(x) &\leq p_u \left(\frac{\lambda x}{(1-\rho)^2} \right) \\
\mathbb{E}[\Delta Res_1(S)] &\leq p_u \left(\frac{\rho}{(1-\rho)^2} \right)
\end{aligned}$$

Therefore, $C_{\lambda, S} = \frac{\rho}{(1-\rho)^2}.$

6.4 Bounding Expected Second Res Time Difference

Main Lemma and Upper Bound

Goal: Prove the following lemma:

$$\text{Lem: } \lim_{p_k \rightarrow 1} E[\Delta Res_2(S)] = 0$$

Steps: 1. Show that the term goes to zero for all x :

$$\begin{aligned}
\lim_{p_k \rightarrow 1} \left(\Delta Res_2(x) - p_u(x) \frac{x}{1-\rho_x} \right) &\leq 0, \quad \forall x \\
\Delta Res_2(x) &= p_u(x) \left(\frac{x}{1-\rho_x} - \int_{t=0}^x \frac{1}{1-\rho_t} dt \right) \\
&\leq p_u(x) \frac{x}{1-\rho_x}
\end{aligned}$$

Note: $\lim_{p_k \rightarrow 1} \left(\frac{x}{1-\rho_x^k} - \frac{x}{1-\rho_x} \right) = 0$ (proof follows from $\Delta Res_1(x)$ argument).

2. Show that the expectation goes to zero:

$$\begin{aligned} \lim_{p_k \rightarrow 1} E \left[p_u(S) \cdot \frac{S}{1-\rho_S} \right] &= 0 \\ E \left[\frac{S_u}{1-\rho_{S_u}} \right] \cdot p_u &= \int_{x=0}^{\infty} p_u(x) \frac{x}{1-\rho_x} f_{S_u}(x) dx \end{aligned}$$

The Maximizing Distribution

For any S distribution and p_k value, there exists an S_u distribution that maximizes $E \left[p_u(S) \cdot \frac{S}{1-\rho_S} \right]$.

The maximizing S_u distribution is defined as:

$$\begin{aligned} S_u &\sim [S \mid S \geq t] \quad \text{where } F_S(t) = p_k \\ S_k &\sim [S \mid S < t] \end{aligned}$$

Argument: The function $\frac{x}{1-\rho_x}$ is increasing and does not depend on S_u . Thus, assigning larger values to S_u leads to larger expectations.

Proof of Limit using the Maximizing Distribution

We define the worst-case distribution $S_u^* := [S \mid S \geq t]$. We want to show:

$$\lim_{p_u \rightarrow 0} p_u E \left[\frac{S_u^*}{1-\rho_{S_u^*}} \right] = 0$$

Where the probability of a job being type u under this policy is:

$$p_u^*(x) = \begin{cases} 1 & \text{if } x \geq t \\ 0 & \text{otherwise} \end{cases}$$

Setting up the integral:

$$\begin{aligned} \int_{x=0}^{\infty} p_u f_{S_u^*}(x) \frac{x}{1-\rho_x} dx &= \int_{x=0}^{\infty} p_u^*(x) f_S(x) \frac{x}{1-\rho_x} dx \\ &= \int_{x=t}^{\infty} f_S(x) \frac{x}{1-\rho_x} dx \end{aligned}$$

To solve this, we use the substitution $z := \rho_x$:

$$\begin{aligned} z &:= \rho_x = \lambda E[S \mathbb{I}(S \leq x)] = \lambda \int_{y=0}^x y f_S(y) dy \\ \frac{dz}{dx} &= \lambda x f_S(x) \end{aligned}$$

Applying the substitution to our integral:

$$\begin{aligned} \int_{z=\rho_t}^{\rho_\infty=\rho} \frac{\frac{1}{\lambda} dz}{1-z} &= \frac{1}{\lambda} \left[-\ln(1-z) \right]_{z=\rho_t}^{\rho} \\ &= \frac{1}{\lambda} (\ln(1-\rho_t) - \ln(1-\rho)) \\ &= \frac{1}{\lambda} \ln \left(\frac{1-\rho_t}{1-\rho} \right) \end{aligned}$$

Finally, evaluating the limit as $p_u \rightarrow 0$:

$$\begin{aligned}\lim_{p_u \rightarrow 0} \rho_t &= \rho \\ \lim_{p_u \rightarrow 0} \frac{1 - \rho_t}{1 - \rho} &= 1 \\ \lim_{p_u \rightarrow 0} \ln \left(\frac{1 - \rho_t}{1 - \rho} \right) &= 0\end{aligned}$$

6.5 Bounds for $\rho - \rho_t$

Lower bound

We approximate the partial load difference:

$$\begin{aligned}\ln(1 + p_u \cdot t \cdot c) \\ \frac{1 - \rho_t}{1 - \rho} &= 1 + p_u \cdot t \cdot c \\ \frac{(1 - \rho) + (\rho - \rho_t)}{1 - \rho} &= 1 + \frac{\rho - \rho_t}{1 - \rho}\end{aligned}$$

Claim: $\rho - \rho_t = p_u \cdot t \cdot c$

Using the integral definition of the partial load difference and tail probability:

$$\rho - \rho_t = \lambda \int_{x=t}^{\infty} x f_S(x) dx \quad (13)$$

$$p_u = \int_{x=t}^{\infty} f_S(x) dx = \bar{F}_S(t) \quad (14)$$

Since $x \geq t$ over the domain of integration, we establish a strict lower bound:

$$\rho - \rho_t \geq p_u \cdot t \cdot \lambda \quad (15)$$

Conditional Expectation

To establish the upper bound, we evaluate the conditional expectation $E[X \mid X > t]$:

$$E[X \mid X > t] = \frac{\int_{x=t}^{\infty} x f_S(x) dx}{P(X > t)} \quad (16)$$

Suppose $E[X \mid X > t] = p \cdot t$ for some constant $p > 1$. Substituting this assumption yields:

$$\int_{x=t}^{\infty} x f_S(x) dx = p \cdot t \cdot P(X > t) \quad (17)$$

Differentiating both sides with respect to t :

$$\begin{aligned}-t f_S(t) &= -p \cdot t \cdot f_S(t) + p P(X > t) \\ (p - 1)t f_S(t) &= p P(X > t)\end{aligned}$$

Isolating the hazard rate yields the exact form for a Pareto distribution:

$$\frac{f_S(t)}{P(X > t)} = \left(\frac{p}{p - 1} \right) \frac{1}{t} \quad (18)$$

Final Upper Bound

We can now rewrite the partial load difference exactly in terms of the conditional expectation:

$$\rho - \rho_t = p_u \cdot E[S \mid S > t] \cdot \lambda \quad (19)$$

As a result, for any distribution S with a finite mean ($E[S] < \infty$), we can bound $E[S \mid S > t] \leq t \cdot C$, establishing the final upper bound:

$$\rho - \rho_t \leq p_u \cdot t \cdot \lambda \cdot C \quad (20)$$

To establish the upper bound, we first examine the general derivative of the conditional expectation $E[X \mid X > t]$ with respect to t . By definition:

$$E[X \mid X > t] = \frac{\int_{x=t}^{\infty} x f_S(x) dx}{P(X > t)} \quad (21)$$

Applying the quotient rule, we differentiate both sides with respect to t :

$$\frac{d}{dt} E[X \mid X > t] = \frac{-t f_S(t) P(X > t) - \left(\int_{x=t}^{\infty} x f_S(x) dx \right) (-f_S(t))}{P(X > t)^2} \quad (22)$$

$$= -\frac{t f_S(t)}{P(X > t)} + \frac{f_S(t)}{P(X > t)} \left(\frac{\int_{x=t}^{\infty} x f_S(x) dx}{P(X > t)} \right) \quad (23)$$

Recognizing the second term inside the parentheses is our original definition of $E[X \mid X > t]$, we can factor out the hazard rate to get a fundamental differential equation:

$$\frac{d}{dt} E[X \mid X > t] = \left(E[X \mid X > t] - t \right) \frac{f_S(t)}{P(X > t)} \quad (24)$$

To understand the worst-case tail behavior, we consider a bounding case where the conditional expectation grows linearly, such that $E[X \mid X > t] = p \cdot t$ for some constant $p > 1$. In this scenario, the derivative is simply p . Substituting this into our differential equation yields:

$$p = (p \cdot t - t) \frac{f_S(t)}{P(X > t)} \quad (25)$$

Isolating the hazard rate yields the exact form for a Pareto distribution:

$$\frac{f_S(t)}{P(X > t)} = \left(\frac{p}{p-1} \right) \frac{1}{t} \quad (26)$$

Because system stability requires the overall mean to be finite ($E[X] < \infty$), the conditional expectation is bounded above by this linear growth rate. Thus, we can bound $E[X \mid X > t] \leq C \cdot t$ for some constant C .

6.6 Distribution-Specific Convergence Rates

Exponential Distribution Analysis

Exponential Distribution Setup

$$\begin{aligned} S &\sim \text{Exp}(\mu) \\ F_S(t) &= p_k \\ \bar{F}_S(t) &= p_u \end{aligned}$$

Exponential: $F_{\text{Exp}}(x) = 1 - e^{-\mu x}$
 $\bar{F}_{\text{Exp}(\mu)}(x) = e^{-\mu x}$
 $p_u = e^{-\mu t} \implies t = \frac{-\ln(p_u)}{\mu}$

Calculating the partial load ρ_t

$$\begin{aligned}\rho_t &= \lambda \int_{x=0}^{x=t} x f_S(x) dx \\ &= \lambda \int_{x=0}^{x=t} x \mu e^{-\mu x} dx \\ &= \lambda \mu \int_{x=0}^{x=t} x e^{-\mu x} dx\end{aligned}$$

(Side note on the antiderivative: $\frac{d}{dx} \left(\frac{-x e^{-\mu x}}{\mu} - \frac{e^{-\mu x}}{\mu^2} \right) = x e^{-\mu x}$)

Evaluating the integral:

$$\begin{aligned}\rho_t &= \lambda \mu \left[\frac{-x e^{-\mu x}}{\mu} - \frac{e^{-\mu x}}{\mu^2} \right]_{x=0}^{x=t} \\ &= \lambda \mu \left[\frac{1}{\mu^2} - \frac{t e^{-\mu t}}{\mu} - \frac{e^{-\mu t}}{\mu^2} \right] \\ &= \frac{\lambda}{\mu} - \lambda t e^{-\mu t} - \frac{\lambda}{\mu} e^{-\mu t}\end{aligned}$$

Convergence bound analysis

Evaluating $\ln \left(\frac{1-\rho_t}{1-\rho} \right)$:

$$\begin{aligned}\ln \left(\frac{1-\rho + \lambda t e^{-\mu t} + \frac{\lambda}{\mu} e^{-\mu t}}{1-\rho} \right) &= \ln \left(1 + \frac{(\lambda t + \frac{\lambda}{\mu}) e^{-\mu t}}{1-\rho} \right) \\ &\approx \frac{(\lambda t + \frac{\lambda}{\mu}) e^{-\mu t}}{1-\rho} \\ &= \frac{p_u (\lambda t + \frac{\lambda}{\mu})}{1-\rho} \\ &\leq p_u \cdot \ln(p_u) \cdot C\end{aligned}$$

(where $C = \frac{\lambda}{1-\rho} + e$)

Note: not linear convergence.

Bounding Expected Wait Time (Pareto)

General Bound Condition

If $\bar{F}_S(x) \leq e^{-cx}$ for all large enough x
and $f_S(x) \leq e^{-cx}$ for all large enough x ,
then $\mathbb{E}[\Delta \text{Wait}(S)] \leq C \cdot p_u \cdot \ln(p_u)$

Pareto Distribution Setup

$$\begin{aligned}S &\sim \text{Pareto}(\alpha, x_m = 1) \\ \bar{F}_S(x) &= x^{-\alpha} \\ f_S(x) &= \frac{\alpha}{x^{\alpha+1}}\end{aligned}$$

$$\begin{aligned}\bar{F}_S(t) &= p_u \\ t^{-\alpha} = p_u &\implies t = (p_u)^{-\frac{1}{\alpha}}\end{aligned}$$

Calculating ρ_t for Pareto

$$\begin{aligned}\rho_t &= \lambda \int_{x=1}^{x=t} x f_S(x) dx \\ &= \lambda \int_{x=1}^{x=t} x \cdot \frac{\alpha}{x^{\alpha+1}} dx \\ &= \lambda \int_{x=1}^{x=t} \frac{\alpha}{x^\alpha} dx \\ &= \lambda \left[\frac{\alpha}{1-\alpha} \cdot \frac{1}{x^{\alpha-1}} \right]_{x=1}^{x=t} \\ &= \lambda \left(\frac{\alpha}{\alpha-1} - \frac{\alpha}{\alpha-1} \left(\frac{1}{t^{\alpha-1}} \right) \right) \\ &= \lambda \frac{\alpha}{\alpha-1} \left(1 - \frac{1}{t^{\alpha-1}} \right) \\ &= \rho \left(1 - \frac{1}{t^{\alpha-1}} \right)\end{aligned}$$

(Recognizing that $\mathbb{E}[S] = \frac{\alpha}{\alpha-1}$, this simplifies to ρ_t)

Convergence bound for Pareto

$$\begin{aligned}\ln \left(\frac{1-\rho_t}{1-\rho} \right) &= \ln \left(\frac{1-\rho \left(1 - \frac{1}{t^{\alpha-1}} \right)}{1-\rho} \right) \\ &= \ln \left(\frac{1-\rho + \rho \frac{1}{t^{\alpha-1}}}{1-\rho} \right) \\ &= \ln \left(1 + \frac{\rho}{1-\rho} \cdot \frac{1}{t^{\alpha-1}} \right)\end{aligned}$$

$$\begin{aligned}\ln \left(1 + \frac{\rho}{1-\rho} \cdot \frac{1}{t^{\alpha-1}} \right) &\approx \frac{\rho}{1-\rho} \frac{1}{t^{\alpha-1}} \\ &= C \cdot p_u \cdot t \\ &= C \cdot p_u^{1-\frac{1}{\alpha}}\end{aligned}$$

Claim: $\mathbb{E}[\Delta \text{Wait}(S)] \approx C \cdot p_u \cdot t$

Investigating the Moment Integral

$$\int x \mathbb{E}[S_x^2] dx$$

Bounding with $\min(X, Y)$

$P(\min(X, Y) \geq t) = P(X \geq t)$ and $P(Y \geq t) = \bar{F}_S(t)^2$

For $S \sim \text{Pareto}(\alpha, x_{\min} = 1)$:

$$\bar{F}_S(x) = x^{-\alpha}$$

$$\bar{F}_{\min}(x) = x^{-2\alpha}$$

$$\iint \min(x, y)^2 dx dy$$

Expected Value Cutoffs

$$\mathbb{E}[X^3 \mathbb{I}(X \leq Y) + XY^2 \mathbb{I}(Y \leq X)]$$

$$\mathbb{E}[X^3 \mathbb{I}(X \leq Y) + Y^3 \mathbb{I}(Y \leq X)] = \mathbb{E}[\min(X, Y)^3]$$

Finite only if $2\alpha < 3 \implies \alpha < \frac{3}{2}$

Evaluating Specific Expected Value

$$\mathbb{E}[XY^2 \mathbb{I}(Y \leq X)] = \int_{x=0}^{\infty} \int_{y=1}^{y=x} xy^2 \frac{\alpha}{x^{\alpha+1}} \frac{\alpha}{y^{\alpha+1}} dy dx$$

$$= \int_{x=0}^{\infty} \frac{\alpha}{x^{\alpha}} \int_{y=1}^x \frac{\alpha}{y^{\alpha+1}} dy dx$$

Inner Integral Evaluation

Solving the inner integral

$$\left[-\frac{\alpha}{\alpha-2} y^{-(\alpha-2)} \right]_{y=1}^x = \frac{\alpha}{\alpha-2} - \frac{\alpha}{\alpha-2} x^{-(\alpha-2)}$$

$$= \frac{\alpha}{\alpha-2} \left(1 - x^{-(\alpha-2)} \right)$$

Conclusion

$\alpha = \frac{3}{2}$ is true cutoff